The equations of motion

AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.8 | Sections 2.8

Goal: Choose and use the right suvat equation for uniform acceleration in a straight line.

01 The big picture

For uniform (constant) acceleration in a straight line, five quantities describe the motion: s, u, v, a and t. Each equation links four of them, so if you know three you can find the others.

Pick the equation that leaves out the quantity you neither know nor need. Choose a positive direction and give every vector (s, u, v, a) a sign.

suvat equations [paraphrase]
The four equations of uniformly accelerated motion in a straight line.

02 Key equations

No s
$$v = u + at$$

No a
$$s = \tfrac{1}{2}(u + v)\,t$$

No v
$$s = ut + \tfrac{1}{2}at^2$$

No t
$$v^2 = u^2 + 2as$$

Symbols and units

s	displacement	m	u, v	initial, final velocity	$\mathrm{m\,s^{-1}}$	a	acceleration (constant)	$\mathrm{m\,s^{-2}}$

03 Sample problems with solutions

Sample 1 · Braking distance

A car travelling at 25 m s^{-1} brakes with a constant deceleration of 5.0 m s^{-2}. Calculate its stopping distance.

Step 1. Known: $u = 25$, $v = 0$, $a = -5.0$.
Unknown: s. No t, so use $v^2 = u^2 + 2as$:

$$0 = 25^2 + 2(-5.0)\,s$$

Step 2. Rearrange for s:

$$s = \frac{625}{10} = 62.5\ \mathrm{m}$$

Answer: $s = 63\ \mathrm{m}$ (2 s.f.)

Unit check: $\frac{(\mathrm{m\,s^{-1}})^2}{\mathrm{m\,s^{-2}}} = \mathrm{m}$ ✓

04 Practice problems

P1 A car accelerates from rest at 2.0 m s^{-2} for 8.0 s. How far does it travel?
Answer: 64 m

P2 A train speeds up from 10 m s^{-1} to 30 m s^{-1} over 400 m. Find its acceleration.
Answer: 1.0 m s^{-2}

P3 A ball is thrown straight up at 15 m s^{-1}. Find its maximum height ($g = 9.81$ m s^{-2}).
Answer: 11.5 m (11 m)

P4 A cyclist at 4.0 m s^{-1} accelerates at 1.5 m s^{-2} for 6.0 s. Find the final velocity.
Answer: 13 m s^{-1}

05 Other useful information

Common mistakes

✗ **Wrong:** Make a positive when the object slows down.
✓ **Right:** If u is positive and it slows, a is negative.

Exam tips

- Write “s = ?, u = ..., v = ..., a = ..., t = ...” first; the empty slot picks the equation.

What the syllabus asks

- 9702 topic 2: solve problems with the suvat equations. Coursebook 2.8.

Explore: Practice questions for this lesson